Ryley McConkey PhD, P.Eng.

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Google Scholar | Personal Website | LinkedIn | GitHub

EDUCATION

2020–2024 **PhD in Mechanical and Mechatronics Engineering**
University of Waterloo, Canada
Thesis: *Machine Learning Methods for Turbulence Closure Modelling*
Supervisors: Fue-Sang Lien, Eugene Yee

2022 **Visiting PhD Student** (6 months)
University of Manchester, UK
Project: Machine Learning-Augmented Simulation of Massively Separated 3D Flows
Supervisors: Alistair Revell, Alex Skillen

2019–2020 **Master of Applied Science in Mechanical Engineering** (direct transfer to PhD)
University of Waterloo, Canada
Thesis: *Numerical Simulation of Turbulent Vortex-Induced Vibration*
Supervisors: Fue-Sang Lien, William Melek

2014–2019 **Bachelor of Science in Mechanical Engineering** (Co-op Program), with Distinction
University of Alberta, Canada
Cumulative GPA 3.9/4.0; ranked 4th of 229

PROFESSIONAL LICENSURE

2024–Present **Professional Engineer (P.Eng.)**, Professional Engineers Ontario

RESEARCH INTERESTS

- Computational fluid dynamics
- Fluid mechanics, aerodynamics, and geophysical fluid dynamics
- Machine learning for computational fluid dynamics and turbulence modelling

RESEARCH AND PROFESSIONAL EXPERIENCE

2024–Present **Postdoctoral Fellow**, Massachusetts Institute of Technology
PIs: Tess Smidt and Abigail Bodner
Affiliations: Earth, Atmospheric and Planetary Sciences (EAPS); Center for Computational Science and Engineering (CCSE); Research Laboratory of Electronics (RLE); Laboratory for Information and Decision Systems (LIDS)

Equivariant machine learning for fluid and ocean modelling

- Designing equivariance-preserving machine learning architectures for problems in physics, with a focus on fluid mechanics
- Built the machine learning data pipeline for a 500 TB ocean simulation dataset on an HPC system, reducing rendering time from approximately 7 hours to under 10 seconds
- Applying models to numerical simulation in ocean modelling, turbulence modelling, and aerodynamics, including work with differentiable Python-based CFD codes
- Initiated the collaboration bridging equivariant machine learning and ocean modelling that produced several of the publications below

Machine learning for constitutive modelling

- Lead the machine learning effort within CHEFSI, a DOE/NNSA PSAAP-IV Predictive Simulation Center, directing a team of 4–5 graduate students and one postdoctoral researcher
- Learning material response for thermal protection systems under atmospheric re-entry conditions
- Set technical direction, run weekly meetings, and represent the workstream externally
- Translate between solid mechanics, materials science, and machine learning groups across the center

2023	**Research Scientist**, Rowan Williams Davies & Irwin (RWDI) Consulting

- Applied PhD techniques to real-world simulation problems at a major Canadian wind engineering firm, reaching accuracy comparable to large eddy simulation at 1.5–2 times the cost of a baseline RANS simulation
- Adapted academic techniques to work with real-world data
- Implemented a custom OpenFOAM (C++) solver interfacing with a Python-based machine learning pipeline
- Communicated research results and complex techniques to varied audiences

2020–2024 **Doctoral Researcher**, University of Waterloo

- Used machine learning models to replace traditional eddy viscosity models in RANS simulations
- Implemented an end-to-end data and machine learning pipeline: data generation from numerical simulation, preprocessing, model training, hyperparameter optimization, and evaluation on high-performance GPU systems
- Substantially modified OpenFOAM turbulence models and solver code to incorporate machine learning predictions
- Supplementary projects: Bayesian optimization of RANS model coefficients (turbo-RANS), simulation of aircraft ice accretion, aeroacoustics of vortex-induced vibration, unsupervised clustering of wake flow patterns, and small modular reactor (SMR) modelling with a 1D systems code

2021–2022 **Software Developer**, RWDI / Orbital Stack

- Built automated simulation workflows with custom OpenFOAM-based wind engineering scripts
- Proposed and demonstrated new machine learning algorithms within existing service lines, including flow field reconstruction from experimental measurements, building energy usage estimation, and experimental data augmentation
- Led evaluation and testing of a new machine learning service line (Orbital Stack AI), a surrogate model for traditional CFD simulation

2019–2021 **Simulation and Design Engineer**, MACH32

- Designed lifesaving autoinjectors for in-situ emergency trauma treatment
- Conducted engineering analyses of design components using CFD
- Invented a negative-pressure containment device in response to hospital staff needs during the COVID-19 pandemic
- Named inventor on five patent applications

2019–2020 **Master's Researcher**, University of Waterloo

- Analyzed and optimized a wind energy harvesting device based on extracting power from vortex-induced vibrations
- Used OpenFOAM extensively, including implementing new time-integration schemes in fluid-structure interaction (FSI) solvers
- Employed deforming mesh, overset mesh, and immersed boundary methods for FSI

2018–2019 **Undergraduate Researcher**, University of Alberta

- Used STAR-CCM+ to simulate flow over an airfoil and predict lift and drag in ground effect

2017–2018 **Turnaround Co-op Student**, Imperial Oil (12 months)

- Conducted engineering analyses, managed fabrication of new equipment, and developed mechanical and piping repair specifications for the 2018 Spring Turnaround
- Created drawings and designs for piping modifications, welding repairs, and pressure vessel alterations

2016–2017 **Project Management Co-op Student**, Enbridge Pipelines (8 months)

- Analyzed cost, risk, and code compliance for large pipeline projects

2016 **Undergraduate Researcher**, University of Alberta

- Studied welding heat transfer by comparing theoretical predictions to experimental results from a FANUC welding robot

TEACHING EXPERIENCE AND DEVELOPMENT

2025 **Kaufman Teaching Certificate Program**, Massachusetts Institute of Technology

Winter 2024 **Sessional Instructor**, University of Waterloo
ME 351: Fluid Mechanics (97 students)

- Managed a team of 3 teaching assistants to deliver a core engineering science course
- Redesigned course tutorials and project to incorporate group-based and experiential learning activities
- Proposed and organized a field trip to an external experimental facility
- Achieved a mean instructor rating of **4.9/5.0** (56 student responses)
- 100% of respondents agreed the course created an inclusive learning environment

2024 **Institute for Engineering Teaching Program**, Canadian Engineering Education Association
Affiliated with the Canadian Engineering Accreditation Board

2023 **Pathways for Addressing Disclosures of Racism**, University of Waterloo

2023 **Teaching Assistant**, University of Waterloo
MTE 202: Ordinary Differential Equations

2022 **Teaching Assistant**, University of Waterloo
MTE 202: Ordinary Differential Equations

2021 **Fundamentals of University Teaching Program**, University of Waterloo

2021 **Teaching Assistant**, University of Waterloo
ME 564: Aerodynamics

2020 **Teaching Assistant**, University of Waterloo
ME 564: Aerodynamics

RESEARCH SUPERVISION AND MENTORSHIP

2025 **Certificate in Research Mentorship**, Massachusetts Institute of Technology

2024–Present **Massachusetts Institute of Technology** (7 students: 1 Master's, 6 undergraduate)

- Machine learning for constitutive modelling within the PSAAP-IV center (Master's)
- Representation learning for hydrostatic and non-hydrostatic turbulence
- Temporal inpainting of Surface Water and Ocean Topography (SWOT) data
- Equivariant machine learning of stochastic subgrid scale closure models for quasi-geostrophic ocean flows
- Investigating implicit data augmentation in 3D isotropic and anisotropic turbulence (summer internship)
- Superresolution of 2D turbulence with equivariance as an inductive bias
- Stochastic rounding for Lattice Boltzmann Method simulations with single precision floating point representations

2023 **University of Manchester** (2 students)
Machine learning-based correction of discretization errors in Rayleigh–Bénard convection

2023–2024 **University of Waterloo** (2 students: 1 Master's, 1 undergraduate)

- Kolmogorov–Arnold networks for turbulence modelling, Bayesian calibration of the GEKO model, and 1D systems-code development (Master's; first author on a resulting journal article)
- Career development and professional mentorship (undergraduate)

2020 **University of Waterloo** (capstone design team)
Designing a vortex-induced vibration energy harvester

PUBLICATIONS

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

[4] **R. McConkey**, J. Balla, E. Hofgard, T. Smidt, A. Bodner, "Rotational equivariance and locality in data-driven subgrid-scale closures," *arXiv*:2607.26850 (2026). Under review, *Computer Methods in Applied Mechanics and Engineering*.

[3] **R. McConkey**, T. Buchanan, T. Smidt, A. Bodner, R. Dwight, P. Cinnella, “The Closure Challenge: a benchmark task for machine learning in turbulence modelling,” *arXiv*:2603.28884 (2026). Under review, Neural Information Processing Systems (NeurIPS).

[2] **R. McConkey**, J. Balla, J. Bailey, A. Backour, E. Hofgard, T. Jaakkola, A. Bodner, T. Smidt, “Turbulence teaches equivariance to neural networks,” *arXiv*:2602.04695 (2026). Under review, *Physical Review Fluids*.

[1] N. Kalia, **R. McConkey**, E. Yee, F. S. Lien, “Bayesian optimization of the GEKO turbulence model for predicting flow separation over a smooth surface,” *arXiv*:2502.11218 (2025).

PEER-REVIEWED JOURNAL ARTICLES

[9] **R. McConkey**, N. Kalia, E. Yee, F. S. Lien, “Realisability-informed machine learning for turbulence anisotropy mappings,” *Journal of Fluid Mechanics* 1019, A49 (2025).

[8] N. Kalia, **R. McConkey**, E. Yee, F. S. Lien, “Kolmogorov–Arnold networks for turbulence anisotropy mapping,” *Physics of Fluids* 37, 085140 (2025).

[7] **R. McConkey**, N. Kalia, E. Yee, F. S. Lien, “Turbo-RANS: straightforward and efficient Bayesian optimization of turbulence model coefficients,” *International Journal of Numerical Methods for Heat and Fluid Flow* 34(8), 2986–3016 (2024).

[6] **R. McConkey**, E. Yee, F. S. Lien, “On the generalizability of machine-learning-assisted anisotropy mappings for predictive turbulence modelling,” *International Journal of Computational Fluid Dynamics* 36(7), 555–577 (2022).

[5] Z. Cheng, **R. McConkey**, E. Yee, F. S. Lien, “Numerical investigation of noise suppression and amplification in forced oscillations of single and tandem cylinders in high Reynolds number turbulent flows,” *Applied Mathematical Modelling* 117, 652–686 (2023).

[4] M. Cann, **R. McConkey**, F. S. Lien, W. Melek, E. Yee, “A data-driven approach for generating vortex-shedding regime maps for an oscillating cylinder,” *Energies* 16(11), 4440 (2023).

[3] **R. McConkey**, E. Yee, F. S. Lien, “Deep structured neural networks for turbulence closure modelling,” *Physics of Fluids* 34, 035110 (2022).

[2] Y. Wu, Z. Cheng, **R. McConkey**, F. S. Lien, E. Yee, “Modelling of flow-induced vibration of bluff bodies: a comprehensive survey and future prospects,” *Energies* 15(22), 8719 (2022).

[1] **R. McConkey**, E. Yee, F. S. Lien, “A curated dataset for data-driven turbulence modelling,” *Nature Scientific Data* 8, 255 (2021).

PATENT APPLICATIONS

[5] M. Curial, C. Terriff, W. Comeau, W. Comeau, **R. McConkey**, “Devices, systems and methods for medicament delivery,” World Intellectual Property Organization, No. WO2023077225A1 (2022).

[4] M. Curial, C. Terriff, W. Comeau, W. Comeau, **R. McConkey**, “Devices, systems and methods for medicament delivery,” World Intellectual Property Organization, No. WO2021087607A1 (2020).

[3] M. Curial, C. Terriff, W. Comeau, **R. McConkey**, “Devices, systems and methods for medicament delivery,” U.S. Patent No. US-20220370718-A1 (2022).

[2] M. Curial, C. Terriff, W. Comeau, **R. McConkey**, “Portable negative pressure isolation unit,” U.S. Patent No. US-20220104982-A1 (2022).

[1] M. Curial, C. Terriff, W. Comeau, B. Koravankudi, W. Comeau, **R. McConkey**, “IMSAFE: a novel large-volume intramuscular autoinjector,” World Intellectual Property Organization, PCT No. PCT/CA2022/050057 (2022).

INVITED TALKS

[5] **R. McConkey**, “Machine learning for turbulence modelling,” Community Seminar, Center for Computational Science and Engineering, Massachusetts Institute of Technology (2026).

[4] **R. McConkey**, “Accelerating fluid simulations with machine learning,” Black Hole Initiative, Harvard University (2025).

[3] **R. McConkey**, A. Backour, J. Balla, E. Hofgard, J. Nigam, T. Smidt, “Rotational equivariance as an inductive bias in machine learning for fluids,” Data Science and Artificial Intelligence Seminar Series, Chalmers University of Technology (2025).

[2] **R. McConkey**, “Turbulence modelling using machine learning: key challenges and recent progress,” Modelling and Simulation Seminar Series, University of Manchester (2022).

[1] **R. McConkey**, “Hands-on data-driven turbulence modelling with PyTorch,” Modelling and Simulation Seminar Series, University of Manchester (2022).

CONFERENCE PAPERS, POSTERS, AND PRESENTATIONS

[13] J. Balla, J. Bailey, A. Backour, E. Hofgard, T. Jaakkola, T. Smidt, **R. McConkey** (presenter), “Implicit augmentation from distributional symmetry in turbulence super-resolution,” Machine Learning and the Physical Sciences, NeurIPS, San Diego, USA (2025).

[12] **R. McConkey** (presenter), J. Balla, E. Hofgard, T. Smidt, “Equivariant machine learning of sub-grid scale closure models for large eddy simulation,” APS Division of Fluid Dynamics Annual Meeting, Houston, USA (2025).

[11] **R. McConkey** (presenter), S. Peng, S. Snider, S. Silvestri, T. Smidt, A. Bodner, “Multi-scale ocean turbulence with Euclidean neural networks,” Gordon Research Conference: Machine Learning for Actionable Climate Science, Rhode Island, USA (2025).

[10] **R. McConkey** (presenter), A. Backour, J. Balla, E. Hofgard, J. Nigam, T. Smidt, “The role of local rotational symmetries and equivariance in data-driven fluid mechanics,” 33rd Annual Conference of the Computational Fluid Dynamics Society of Canada, Montreal, Canada (2025).

[9] **R. McConkey** (presenter), A. Backour, J. Balla, E. Hofgard, J. Nigam, T. Smidt, “On rotational equivariance as an inductive bias in machine learning for fluids,” ERCOFTAC Workshop on Data-Driven Fluid Mechanics, London, UK (2025).

[8] F. S. Lien, E. Yee, D. Orchard, C. Li, Z. Cheng, Y. Wu, H. H. Huang, **R. McConkey** (presenter), J. Wang, L. H. Chen, J. Yi, N. Kalia, “Development of a simulation environment for the assessment of urban air mobility vehicles: energy efficiency, noise, and icing,” Sustainable Aeronautics Summit, Waterloo Institute for Sustainable Aeronautics, Waterloo, Canada (2023).

[7] **R. McConkey** (presenter), A. Mole, A. Skillen, A. Revell, E. Yee, F. S. Lien, “Machine learning augmented turbulence modelling for massively separated three-dimensional flows,” Computational Fluids Conference, Cannes, France (2023).

[6] **R. McConkey**, A. Mole (presenter), A. Skillen, A. Revell, E. Yee, F. S. Lien, “XGBoost-augmented RANS closure modelling of complex 3D flows,” Workshop on Data-Driven Methods for Fluid Mechanics, Leeds Institute for Fluid Dynamics, Leeds, UK (2023).

[5] **R. McConkey**, E. Yee, F. S. Lien, “Deep learning-based turbulence closure with improved optimal eddy viscosity prediction,” Proceedings of the 29th Annual Conference of the Computational Fluid Dynamics Society of Canada (2021).

[4] M. Cann, **R. McConkey**, F. S. Lien, W. Melek, E. Yee, “Mode classification for vortex shedding from an oscillating wind turbine using machine learning,” *Journal of Physics: Conference Series* 2141(1), 012009 (2021).

[3] **R. McConkey**, L. Long, A. Komrakova, J. G. Wong, “Evaluation of RANS turbulence models and boundary conditions for CFD simulations of a finite wing in ground effect,” Okanagan Fluid Dynamics Meeting, Canmore, Canada (2019).

[2] **R. McConkey** (presenter), L. Long, A. Komrakova, J. G. Wong, “3D simulations of a finite wing in ground effect,” Undergraduate Research Symposium, University of Alberta (2019).

[1] **R. McConkey** (presenter), L. Long, A. Komrakova, J. G. Wong, “Simulations of a 2D NACA 0012 airfoil in ground effect,” Undergraduate Research Symposium, University of Alberta (2018).

SOFTWARE AND DATASETS

2025–Present **The Closure Challenge**
Benchmark suite and public leaderboard for machine learning in turbulence modelling, hosted under ERCOFTAC Special Interest Group 54

- Originated the benchmark and serve as benchmark steward: evaluate all submissions, maintain the leaderboard, and set governance policy
- Built the open-source evaluation and scoring package (`closure-challenge` on PyPI), the metrics, and the submission process
- Six leaderboard submissions to date from international research groups

2024–Present	**turbo-rans** Black-box Bayesian calibration of turbulence model coefficients (PyPI, GitHub) • Includes templates for OpenFOAM, ANSYS Fluent, STAR-CCM+, and a solver-agnostic interface
2021–Present	**A Curated Dataset for Data-Driven Turbulence Modelling** Collocated RANS and high-fidelity data for the same flows, structured for immediate use in machine learning • 895,640 data points across five flow families and four turbulence models; the first open-source dataset of its kind • 5,415 Kaggle downloads, with usage still accelerating five years after release
2021–Present	**dataFoam** OpenFOAM-to-dataframe converter with three custom OpenFOAM applications and a unit test suite (GitHub)

AWARDS AND SCHOLARSHIPS

2024–2026	NSERC Postdoctoral Fellowship
2022–2025	NSERC Postgraduate Scholarship (Doctoral)
2023	Mitacs Accelerate Research Internship Award
2022	Ontario Graduate Scholarship (offered, declined)
2021	Best Student Paper, Computational Fluid Dynamics Society of Canada Annual Conference
2021	Computational Fluid Dynamics Society of Canada Graduate Scholarship
2021	Mitacs Globalink Research Award, UK Research and Innovation
2020–2024	University of Waterloo President’s Graduate Scholarship
2020–2022	Ontario Graduate Scholarship
2019	Tyler Lewis Clean Energy Research Foundation Grant
2019	University of Waterloo Engineering Dean’s Entrance Award
2019	University of Alberta Dean’s Research Award for Undergraduate Research
2019	University of Alberta Engineering Economics in Design Award for Capstone Project
2018	University of Alberta Undergraduate Research Conference, Student’s Choice Award
2016	University of Alberta Green and Gold Student Leadership & Professional Development Grant
2016	University of Alberta Dean’s Research Award for Undergraduate Research
2014–2017	University of Alberta Jason Lang Scholarship

TECHNICAL SKILLS

Simulation	**OpenFOAM** at the source level: modified two-equation RANS models, one-equation LES models, and the `simpleFoam`, `pisoFoam`, and `pimpleFoam` solvers to interface with machine learning models; implemented a custom fluid-structure interaction integrator in foam-extend. **ANSYS Fluent and CFX**, **STAR-CCM+**, **ParaView**. Meshing with snappyHexMesh, blockMesh, and Pointwise. Wrote lattice Boltzmann and 1D compressible systems codes from scratch. Experience with Oceananigans, phiFlow, and JAX-CFD.
Machine learning	**PyTorch** including distributed multi-node training on H100 nodes, custom dataloaders, and equivariant architectures (ESCNN). scikit-learn, XGBoost and dask-XGBoost (scaled to 56 million rows across four A100s), JAX, TensorFlow and Keras, Hugging Face, and hyperparameter search.
Languages	**Python** (three public PyPI releases; neural networks and PISO solvers implemented from scratch), **LaTeX**, Bash and Linux, MATLAB, C++, Julia.
Infrastructure	SLURM since 2019 across five HPC systems, Git and GitHub with commercial code-review workflow, Docker, pytest and CI, Python packaging, Azure and Databricks, HDF5, netCDF, xarray, and dask.

Engineering	SolidWorks and Onshape, fabrication and welding design drawings, pressure vessel and piping repair specifications, 3D printing, laser cutting, and machining.

ACADEMIC SERVICE

PEER REVIEW ACTIVITY (2022–PRESENT)

Reviewer for approximately 20 manuscripts and grants:

Journal of Fluid Mechanics • American Institute of Aeronautics and Astronautics (AIAA) • *Physical Review Fluids* • *Nature Machine Intelligence* • *Physics of Fluids* • *Computers and Fluids* • *International Journal of Heat and Fluid Flow* • *International Journal of Computational Fluid Dynamics* • *Aerospace Science and Technology* • Deutsche Forschungsgemeinschaft (DFG)

COMMUNITY AND COMMITTEE ACTIVITY

2026–Present — **Benchmark Steward**, The Closure Challenge
ERCOFTAC Special Interest Group 54
- Evaluate all community submissions, maintain the public leaderboard, and set governance policy for the benchmark

2025–Present — **Organizer**, Machine Learning for Fluids Working Group
Massachusetts Institute of Technology
- Founded and ran a monthly cross-departmental group bringing together researchers working on machine learning for fluid mechanics

2024 — **Graduate Student Representative**, Department Chair Appointment Committee
University of Waterloo
- Sole graduate student representative on the faculty committee
- Gathered graduate student feedback and participated in committee meetings as part of the chair re-nomination procedure

2023–2024 — **Engineering Director**, Board of Directors, Graduate Studies Endowment Fund
University of Waterloo
- Reviewed applications for graduate student initiative funding monthly, collaborated with other board members, and provided funding recommendations

2022–2023 — **Engineering Representative**, Project Review Committee, Graduate Studies Endowment Fund
University of Waterloo

2018–2019 — **Senior Design Lead**, SAE Aero Design Team, University of Alberta

2016–2017 — **President**, Students for Learning High School Tutoring Team, University of Alberta

VOLUNTEERING AND OUTREACH

2026–Present	Volunteer High School Tutor (weekly), Cambridge, Massachusetts
2024	Volunteer Photographer, Canadian Cancer Society Run for the Cure
2023	Volunteer Photographer, Toronto Beaches Jazz Festival
2021	Volunteer Mechanical Design Judge, Canadian Hyperloop Competition
2021	Volunteer Robot Design Judge, First Robotics Canada (STEM outreach)